Muhamad Irfan Afriyansyah

AI Engineer | Machine Learning | RAG and LLM Applications

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Professional Summary

AI Engineer focused on applied machine learning, retrieval-augmented generation, and LLM applications. Builds inspectable AI systems with Python, FastAPI, vector retrieval, and structured generation. Transitioning from data annotation into AI engineering, with Qalbu as the primary case study for safety checks, retrieval, citations, and honest fallback behavior. Interested in AI Engineer, LLM Engineer, and GenAI Engineer roles.

Technical Skills

Languages: Python, JavaScript
AI and LLM: retrieval-augmented generation (RAG), vector search, Jina embeddings (768 dimensions), Gemini structured JSON
Backend and data: FastAPI, Supabase pgvector, Server-Sent Events (SSE)
Reliability and testing: safety guardrails, citation and provenance validation, quality gates, fallback handling, Pytest, API/retrieval/response-quality tests

Professional Experience

Machine Learning Intern | FlyRank AI

June 2026 - Present | *Remote*

Data Annotator | Thoth AI

June 2025 - Present | *Jakarta, Indonesia* | *On-site*

Robotics Trainer | PT. Racer Robotics Indonesia

October 2022 - April 2023 | *South Tangerang, Banten, Indonesia*

Featured Project

Qalbu - Indonesian Quran Reflection Chat

Personal project | *Work in progress* | *Python, FastAPI, RAG, Jina embeddings, Gemini* | *github.com/Irfandroid/qalbu-rag*

- Implemented 768-dimensional Jina retrieval with a documented 0.32 cosine threshold, parent-level deduplication capped at four sources, citation and quality gates, and sources-only fallbacks when evidence is insufficient.
- Repository includes 52 Pytest test functions covering safety, retrieval, citations, response quality, API behavior, providers, and frontend history.
- Uses the active qalbu-seed-v1 corpus with provenance from the LPMQ Ministry of Religious Affairs portal; indexing and runtime are aligned at 768 dimensions.

Education

Bachelor's Degree in Computer Programming | Universitas Pamulang

January 2019 - May 2024 | *GPA: 3.51*

Certifications

AI for Gen Y / Magang Studi Independen Bersertifikat - Orbit Future Academy; AWS re/Start Batch 10 - Orbit Future Academy; Alibaba Cloud Platform for AI (PAI); Model Studio Fundamentals; The Large World of Language Models; Machine Learning Developer - Digital Talent Scholarship; Data Analysis Using Python - IBM; Data Visualization Using Python - IBM.